

Magnetic anisotropy and ferromagnetic ordering in 3d transition metal adsorbed fluorinated borophene γ -B₈F₄: A first-principles study

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ABSTRACT

In this study, a partially surface fluorinated 2-Pmmn borophene has been proposed as stable 2D structure to explore the emerging magnetic and electronic characteristics induced by 3d transition metal adsorption. Amongst functionalized borophene (B₈F₄)'s α -, β - and γ -configurations, the γ -B₈F₄ identifies as energetically favorable and dynamically stable structure due to the lowest formation energy (−89.89 eV) and having no imaginary phonon modes. Adsorption of 3d transition metal atoms (V, Cr, Mn, Fe and Co) at the most favorable site on γ -B₈F₄, significantly induce the magnetic ordering in nonmagnetic γ -B₈F₄. Followed by higher degree of spin asymmetry, Mn-adsorbed γ -B₈F₄ structure leads to maximum magnetic moment (4.87 μ_B). Spin-orbit coupling enables calculation utilizing force field theorem, which reveals magnetic anisotropy energy (MAE = 1.681 meV) in Mn-adsorbed γ -B₈F₄ structure, demonstrating anisotropy-driven stability of 2D magnetism and defining an out-of-plane easy axis for Mn adsorption. Furthermore, Monte-Carlo simulation based on Heisenberg model estimates that Co-adsorbed γ -B₈F₄ structure exhibits ferromagnetic ordering with significant Curie temperature (34.5 K). These findings establish fluorinated borophene as a stable, adaptable 2D magnetic material platform, providing an efficient pathway toward engineering room-temperature ferromagnetism for potential spintronic applications.

1. Introduction

The discovery of graphene in 2004 by Novoselov et al. and its exceptional carrier mobility $\mu > 2 \times 10^5 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$ has led to trigger the intense research for new two-dimensional (2D) materials having novel functionalities [1,2]. The gapless Dirac dispersion of graphene and its thermal conductivity of around $5300 \text{ W m}^{-1}\text{K}^{-1}$ highlights the potential of 2D materials [3]. Researchers have been exploring the possibility of the next breakthrough 2D material, since the discovery of graphene, which lead to research on silicene [4], borophene [5], germanene [6], stanene [7], phosphorene [8], hexagonal boron nitride (h-BN) [9], MXenes [10] and transition metal dichalcogenides (TMDs) [11], etc.

In the Xenes family of 2D materials, borophene, a monolayer allotrope of boron, has gained considerable attention among researchers, because of its unique polymorphic structures, high metallic conductivity and mechanical flexibility [12]. The strong electron-phonon coupling in borophene, resulting from its low mass density, leads to superconductivity with critical temperature up to 29 K [13,14]. Furthermore, the in-plane stiffness of borophene which also depends on its polymorph,

can reach up to 398 Nm^{-1} , thereby making it a strong rival to graphene [15,16]. The fascinating properties of borophene combined with its low atomic weight, has established borophene as a potential candidate for the future technological applications. However, despite of having fascinating properties, the stability of borophene remains debatable due to conflicting studies by various research groups [15,17,18]. The observed negative frequency in phonon spectrum of 2-Pmmn borophene reveals that free standing borophene may not be stable [19]. For the first time, Mannix et al. have experimentally synthesized buckled 2-Pmmn borophene on the silver Ag(111) surface under ultrahigh vacuum conditions [15]. The laboratory synthesis of borophene without metallic substrate makes experimental production even more challenging. Several stabilization techniques such as surface functionalization, encapsulation, etc. are required due to these stability constraints. Surface functionalization has proved to be one of the most successful technique to enhance borophene's experimental utilization. Surface functionalization of the borophene has granted unique properties, which are akin to graphene such as Dirac cone near Fermi level [19]. It will be interesting to unravel the effect of surface functionalization of

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borophene with fluorine atoms, known as fluorination. Fluorination of borophene may lead to stabilize the borophene lattice and tailor its electronic properties. Fluorination of borophene eliminates the lattice instabilities and dangling bonds are saturated by strong B-F bonding [20,21]. Earlier work by A. Mogulkoc et al. has shown that hydrogenation and fluorination significantly modify the electronic and optical properties of borophene [22]. Borophene and fluorinated borophene are nonmagnetic in nature which limits its applicability for spintronic devices.

Although the Mermin-Wagner theorem states that the ferromagnetic ordering in two-dimensional material is forbidden due to thermal fluctuation at finite temperature [23]. However, the stable magnetic ordering in binary and ternary 2D materials e.g. CrI₃ [24], Cr₂Ge₂Te₆ [25] and Fe₃GeTe₂ [26] etc., at finite temperature has been observed theoretically and experimentally. It has been found that magnetic anisotropy and Spin-orbit coupling (SOC) can stabilize the magnetic ordering in low dimensional system even at finite temperature. The quest of introducing magnetic ordering in nonmagnetic materials emerged due to successfully incorporation of magnetic ordering in non-magnetic materials by doping, adsorption of transition-metal atom [27,28], defects [29] and strain [30] etc. Amongst all these techniques, transition metal (TM) adsorption is one of the prominent technique to introduce magnetic ordering in 2D materials. The adsorption of transition metal atom induces localized magnetic ordering in the system due to the partially filled 3d orbitals of TM atom. It has been observed in the previous studies that the interaction of 3d orbital of transition metal with host material successfully induces the magnetic ordering in the system. Transition metal adsorption has been widely studied on graphene [31], phosphorene [32], silicene [33] and TMDs [34,35], etc. In theoretical modeling of TM-adsorbed systems, standard DFT underestimates the localization of 3d-orbital electrons, which leads to magnetic moment predictions with poor accuracy. The advanced DFT+U method considers strong electron-electron interaction within d-orbitals by inserting an on-site coulomb correction parameter (U). Previous studies have suggested that considering Hubbard U correction with standard DFT, significantly improves the accuracy and provides more reliable magnetic and electronic attributes of the system under consideration. Fluorinated borophene offers several advantages as a host for 3d transition metal adsorption as compared to pristine borophene. The fluorination stabilizes the structure and provides stable platform for TM atom adsorption. The higher electronegativity of fluorine atom can enhance the TM atom bonding. Despite these foreseeable advantages in possible TM atom adsorption on fluorinated borophene, these systems have not been systematically studied, which creates a crucial information gap regarding the interaction of magnetic doping and chemical functionalization. In this study, we address this aspect by investigating the adsorption of 3d transition metals V, Cr, Mn, Fe, and Co on fluorinated borophene utilizing first-principles DFT simulations with Hubbard U corrections.

2. Computational method and details

2.1. Density functional theory

In the present study, first principles calculations have been performed utilizing open-source plane-wave density functional theory (DFT) package Quantum ESPRESSO [36]. The electron exchange correlation effect has been treated within the framework of Generalized Gradient Approximation (GGA) with Perdew-Burke-Ernzerhof (PBE) [37] exchange functional. The interaction between core electrons and ions has been described utilizing ultrasoft pseudopotential to achieve computational efficiency with accuracy. The strong electron correlation in the localized 3d orbitals of transition metal atoms, have been considered through the Hubbard correction (DFT+U). The Hubbard U parameters for transition metals (TMs) V, Cr, Mn, Fe and Co have been obtained by the linear response approach [38]. Specifically, a standard

DFT calculation of transition metal adsorbed fluorinated borophene structure has been performed with some initial guess of U values, to extract the unperturbed Kohn-Sham wavefunctions and ground state spin charge density. The extracted information has been utilized for linear response calculation to estimate the U-values. To account for the effect of surrounding atoms on 3d orbitals, transition metal adsorbed fluorinated borophene has been considered for U value calculation. The van der Waals (vdW) interaction between the adsorbed transition metal atom and fluorinated borophene surface have been incorporated utilizing Grimme DFT+D3 correction. All the calculations have been performed with spin-polarized DFT, to determine the contribution of spin-up and spin-down states on the electronic and magnetic properties. Any artificial interaction between periodic images of the structure has been eliminated by introducing a 20 Å vacuum along the out of plane direction (z-axis), which eliminates the possibility of interaction beyond the adsorbed atom and the structure. A kinetic energy cutoff for charge density expansion has been set to 480 Ry. Brillouin zone sampling has been performed utilizing $6 \times 6 \times 1$ Monkhorst-Pack [39] k-point mesh for the structure optimization, with force convergence threshold below 1×10^{-4} Ry/Bohr on each atom. For total energy and electronic structure calculation, SCF (Self-Consistent Field) calculations have been performed with dense $8 \times 8 \times 1$ Monkhorst-Pack k-point mesh with an energy convergence threshold of 1×10^{-8} Ry. The structure considered for single transition metal adsorption contains 32 boron and 16 Fluorine atoms, where transition metal atom V, Cr, Mn, Fe and Co have been adsorbed on energetically favorable sites. To examine the dynamical stability of the α -, β - and γ -configurations of fluorinated borophene, force constant method has been utilized, which calculates the phonon structure utilizing Phonopy code. The thermodynamic stability of the structure has been evaluated utilizing Ab initio molecular dynamics (AIMD) simulation. The simulation has been performed within the canonical (NVT) ensemble using Nose-Hoover thermostat at temperature 300 K. Structural stability has been assessed from the time evolution of the total energy. Spin resolved charge density have been obtained from the spin polarized calculation, The spin density difference ($\rho_{\uparrow} - \rho_{\downarrow}$) has been evaluated utilizing post processing tool available in Quantum ESPRESSO. Visualization of structural and spin distribution have been achieved through packages such as VESTA [40] and XCryS-Den [41].

2.2. Energy mapping method

The magnetic exchange interactions have been calculated utilizing the total-energy mapping method. All DFT calculations have been performed to extract the exchange parameters by considering $2 \times 2 \times 1$ magnetic supercell, containing four transition-metal atoms. The total energy of distinct magnetic configurations (FM, AFM1, AFM2, and AFM3) have been estimated utilizing DFT. Based on the lattice, three independent exchange parameters have been considered, the nearest neighbor exchange interaction along the x-direction (J_1), the nearest neighbor exchange interaction along the y-direction (J_2) and the next-nearest neighbor exchange interaction along diagonal (J_3). A schematic of containing different magnetic configuration and exchange parameters have been added as Fig. S1.

The magnetic interactions have been modeled using the Heisenberg Hamiltonian [42,43],

$$H = -\sum_{ij} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad (1)$$

where J_{ij} denotes the exchange coupling constant between spins at sites i and j , and $\mathbf{S}_i$ and $\mathbf{S}_j$ represent the spin vector at site i and site j , respectively. Based on the Heisenberg Hamiltonian, the ground state energy of the different magnetic configuration (FM, AFM1, AFM2, and AFM3) can be expressed as,

$$E_{FM} = E_0 - 4S^2(J_1 + J_2 + 2J_3) \quad (2)$$

$$E_{AFM1} = E_0 - 4S^2(-J_1 + J_2 - 2J_3) \quad (3)$$

$$E_{AFM2} = E_0 - 4S^2(J_1 - J_2 - 2J_3) \quad (4)$$

$$E_{AFM3} = E_0 - 4S^2(-J_1 - J_2 + 2J_3) \quad (5)$$

Utilizing Eqs. (2) to (5), J_1 , J_2 and J_3 can be expressed as,

$$J_1 = (E_{AFM1} + E_{AFM3} - E_{AFM2} - E_{FM})/16S^2 \quad (6)$$

$$J_2 = (E_{AFM2} + E_{AFM3} - E_{AFM1} - E_{FM})/16S^2 \quad (7)$$

$$J_3 = (E_{AFM1} + E_{AFM2} - E_{AFM3} - E_{FM})/32S^2 \quad (8)$$

where E_{FM} , E_{AFM1} , E_{AFM2} and E_{AFM3} denote the ground state energy for the FM, AFM1, AFM2, and AFM3 configurations, respectively. E_0 represents the non-magnetic energy contribution in total energy.

2.3. Monte-Carlo simulation

The temperature-dependent magnetic properties have been investigated by atomistic spin dynamics simulation as implemented in the VAMPIRE simulation package, which utilized classical Heisenberg model to consider the magnetic interaction in the system. The simulations have been performed on a two-dimensional rectangular lattice using periodic boundary conditions. Exchange parameters upto 3rd nearest neighbour J_1 , J_2 and J_3 have been considered for long range interaction. To reduce the finite size effect 25 nm x 45 nm x 2.6 nm supercell have been constructed, which contains nearly 1600 magnetic atoms. Monte Carlo simulation with metropolis algorithm, implemented in VAMPIRE code, has been performed to estimate the Curie/Neel's temperature. For each temperature, the system was first equilibrated for 10,000 Monte Carlo steps, followed by an average run of 5000 Monte Carlo steps. The temperature was varied over a wide range to capture the magnetic phase transition. The spin configurations at

selected temperatures have been visualized to identify the type of magnetic state.

3. Results and discussions

3.1. Structure and electronic property of 2Pmmn-borophene and fluorinated borophene

The lattice parameters of fully optimized unit cell of 2-Pmmn borophene containing two boron atoms, have been found as $a = 1.62 \text{ \AA}$, $b = 2.86 \text{ \AA}$, with buckling height, $h = 0.89 \text{ \AA}$, respectively. The structure expands in the y-axis and compresses in x-axis due to fluorination, which slightly changes the lattice parameter value of 'a' from $3.24 \text{ \AA}$ to $3.19 \text{ \AA}$ and 'b' from $5.72 \text{ \AA}$ to $5.78 \text{ \AA}$. The stability assessment utilizing formation energy and phonon band diagram calculations, reveal that the structural changes after fluorination enhance the stability of the structure. The estimated formation energy values have been displayed in Table S1. Amongst all the configurations (α , β and γ), the γ -configuration with formation energy -89.89 eV is energetically more favorable. The absence of negative frequency in phonon diagram of γ -B₈F₄ structure, as displayed in Fig. 1(i), reveals that γ -B₈F₄ structure is dynamically more stable and provides stable foundation for TM atom adsorption.

The thermodynamic stability of the γ -B₈F₄ structure has been evaluated utilizing AIMD molecular dynamics simulation. The fluctuation in total energy of the structure with time at temperature 300 K has been displayed in Fig. 2(c). After initial equilibration state, the total energy fluctuates within a narrow range, which indicates that the system reaches thermodynamics equilibrium and remains stable. The initial and final arrangement of atoms in γ -B₈F₄ structure as displayed in Fig. 2(d), further confirms that no noticeable bond construction and breaking or structural distortion has been observed after the simulation time of 5.9 ps.

The band structure and DOS plot of γ -B₈F₄ structure exhibited in Fig. 2(a) and (b), reveals that γ -B₈F₄ structure is metallic and non-magnetic in its pristine form. There has been notable band crossing at fermi level as observed in the band structure of γ -B₈F₄ structure, which

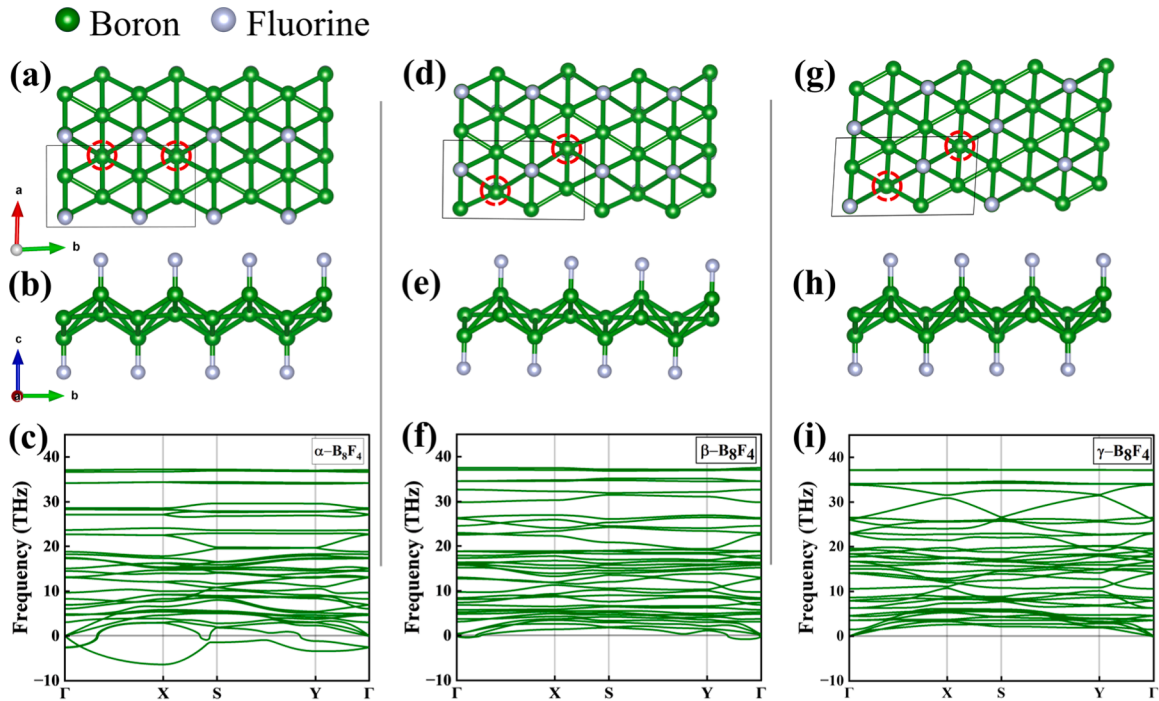


Fig. 1. (a) Top view, (b) Side view and (c) Phonon dispersion diagram for α -B₈F₄, (d) Top view, (e) Side view and (f) Phonon dispersion diagram for β -B₈F₄ and (g) Top view, (h) Side view and (i) Phonon dispersion diagram for γ -B₈F₄ configuration. Black line in top view images denotes the unit cell formation and red circle represents the position of fluorine atom on backside of the structure.

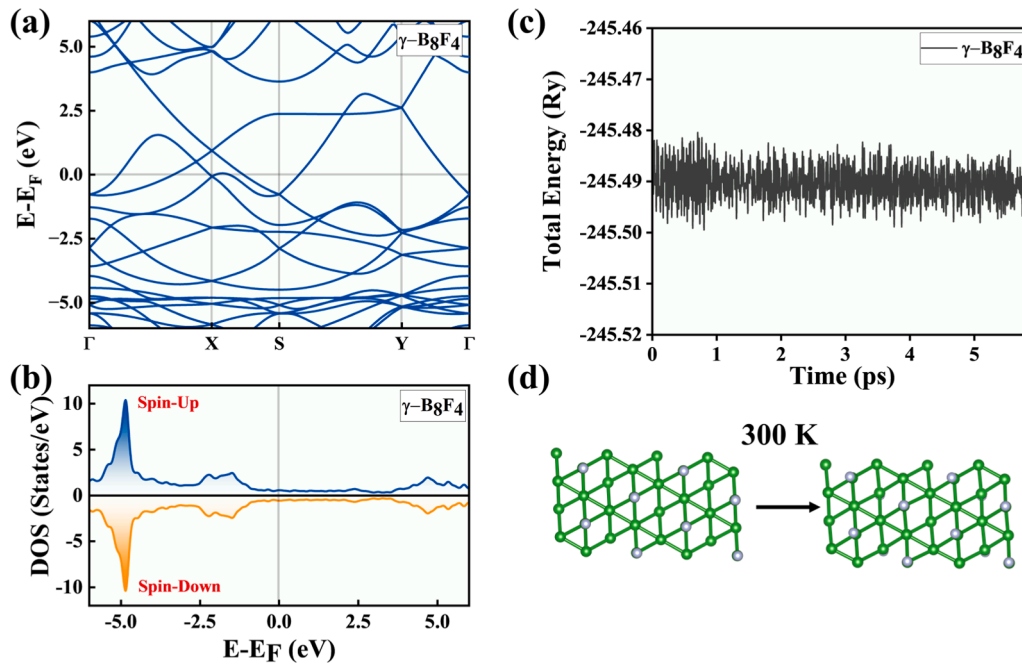


Fig. 2. (a) The band diagram, (b) Total Density of States plot, (c) Energy fluctuation vs time (d) Initial and final structure, during AIMD simulation at 300 K, for fluorinated borophene γ -B₈F₄.

reveals that after halogenation (γ -B₈F₄) structure remains metallic but its electronic states have been significantly modified as compared to pristine 2-Pmmn borophene due to chemical bonding with fluorine atoms. The band and DOS plots of, α -B₈F₄ and β -B₈F₄ have been shown in Fig. S2, which depict that changing fictionization configuration not only effect the stability but also alter the electronic properties. The symmetry in spin-up and spin-down states in DOS, as displayed in Fig. 2(b), exhibits that γ -B₈F₄ structure is nonmagnetic in its pristine form. The nonmagnetic nature of γ -B₈F₄ structure limits its utilization in spintronic and magnetic devices. To overcome this limitation, the adsorption of 3d transition metal atoms (V, Cr, Mn, Fe and Co) on the top surface of the fluorinated borophene, have been considered.

3.2. The structural stability of 3d transition metal adsorbed fluorinated borophene

The 3d transition metal atoms have been adsorbed on two most common sites; the first site is on top of B atom (T_{top}) and the second site is in the middle of fluorine atoms matrices (T_{middle}). The adsorption energy for 3d transition metal atom V, Cr, Mn, Fe and Co have been displayed in Table 1. The adsorption energy has been calculated utilizing the following relation:

$$E_{ads} = E_{Total} - E_{TM} - E_{FB} \quad (1a)$$

Where, E_{Total} is the total energy of 3d transition metal adsorbed fluorinated borophene, E_{TM} is the total energy of isolated 3d transition metal atoms and E_{FB} is the total energy of fluorinated borophene. It has been concluded from Table 1 that at T_{top} site, the adsorption energy is relatively lower for V and Co adsorption as compared to T_{middle} site and for Cr, Mn and Fe, adsorption at T_{middle} site results in lower adsorption energy as compared to T_{top} site.

Table 1

Adsorption energy for 3d transition metal atoms on different adsorption sites of halogenated borophene in eV.

Site	V	Cr	Mn	Fe	Co
T_{top}	-5.10513	-3.31892	-3.60896	-3.48641	-3.69526
T_{middle}	-4.59837	-3.37742	-3.60904	-3.62142	-3.22966

This adsorption energy trend reveals that for V and Co adsorption, T_{top} site is more favorable and for Cr, Mn and Fe adsorption T_{middle} site is more favorable. The respective site for 3d transition metal having minimum adsorption energy has been considered for further study.

The Hubbard U correction has been considered in all the calculations to accurately assess the 3d state of V, Cr, Mn, Fe and Co adsorbed systems. The calculated values of Hubbard parameter U, utilizing linear response method, have been displayed in Table 2. It has been observed from the calculated U values, that the surrounding atoms of 3d transition metal significantly influence the Hubbard parameters. The adsorption of 3d transition metals significantly affects the stability of structure. The formation energy further quantifies the stability of 3d transition metal atom adsorbed structure, which has been estimated utilizing the following relation:

$$E_{formation} = E_{total} - E_{TM} - nE_B - mE_F \quad (2a)$$

Where, E_{total} is the total energy of 3d transition metal adsorbed

Table 2

The Hubbard U parameter, formation energy ($E_{formation}$), charge transfer and magnetic moment of 3d transition metal atom adsorbed fluorinated borophene structure.

TM atom	U (eV)	$E_{formation}$ (eV)	Charge Transfer (e)	Magnetic Moment	
				$M_{isolated}$ (μ_B)	M_{total} (μ_B)
V	6.2575	-367.08	1.43	3 [$3d^3 4s^2$]	2.55
Cr	7.7701	-365.35	1.47	6 [$3d^5 4s^1$]	3.96
Mn	5.6843	-365.58	1.54	5 [$3d^5 4s^2$]	4.87
Fe	6.3095	-365.59	1.24	4 [$3d^6 4s^2$]	3.55
Co	6.0181	-365.67	1.02	3 [$3d^7 4s^2$]	1.75

fluorinated borophene, E_{TM} is the total energy of isolated 3d transition metal atoms and E_B and E_F is the total energy of isolated B and F atom, respectively and ($n = 32$) and ($m = 16$) denotes the respective number of B and F atoms. The formation energy has been calculated per cell and exhibited in Table 2, which elucidate that all the TM doped structures are stable and can be realized experimentally.

3.3. Electronic and magnetic properties of 3d transition metal adsorbed fluorinated borophene

The adsorption of 3d transition metals V, Cr, Mn, Fe and Co significantly alters the electronic and magnetic states of fluorinated

borophene. Pristine (γ -B₈F₄) configuration exhibits metallic behavior with symmetry in spin-up and spin-down states, which indicates that no net magnetic ordering present in the structure as shown in Fig. 2(b). The presence of unpaired electron in the adsorbed system leads to magnetic moment values of 2.55 μ_B , 3.96 μ_B , 4.87 μ_B , 3.55 μ_B and 1.75 μ_B for V-, Cr-, Mn-, Fe- and Co- adsorbed γ -B₈F₄ structure, respectively as exhibited in Table 2. For an isolated transition metal atom, the availability of unpaired electron in outer '3d-orbital' for V, Mn, Fe and Co and '3d and 4s-orbital' for Cr gives rise to net magnetic moment. However, the interaction of these orbitals with surrounding atoms, redistributes the electron in the structure due to crystal field splitting, hybridization and charge transfer. This redistribution of electrons give rise to altered

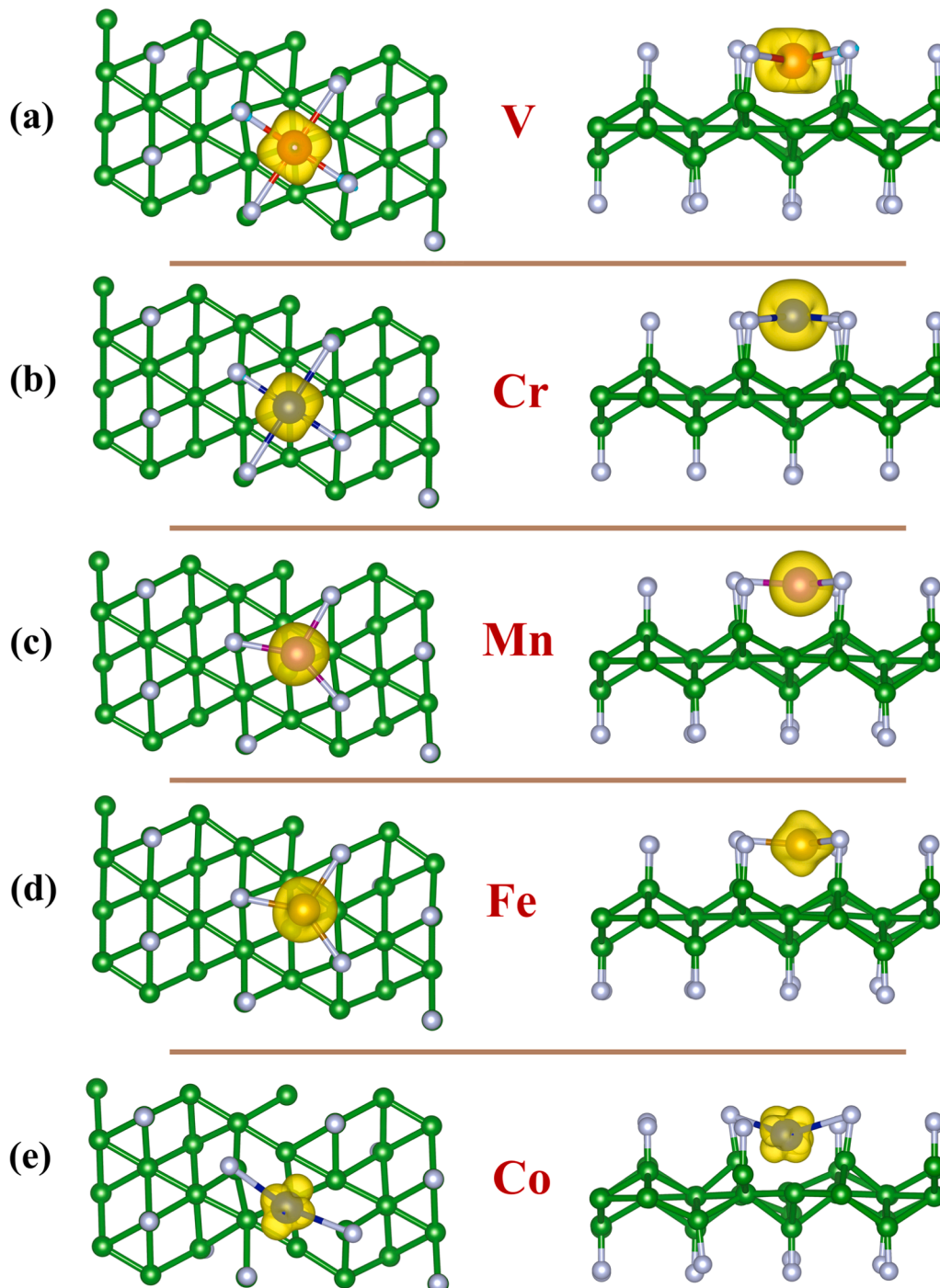


Fig. 3. Spin density difference isosurface of (a) V-, (b) Cr-, (c) Mn-, (d) Fe- and (e) Co-adsorbed structure, having isosurface value 0.0194.

magnetic moment in the system. The isolated V and Co atom exhibits magnetic moment of $3 \mu_B$ but after adsorption the magnetic moment reduces. In V-adsorbed system charge transfer from 3d,4 s orbital reduces the magnetic moment from $3 \mu_B$ to $2.55 \mu_B$, while in Co-adsorbed system, charge transfer is significantly lower as compared to V but magnetic moment decreases to $1.75 \mu_B$. The quenching of magnetic moment also plays critical role to reduce the magnetic moment in these systems.

The spin density difference ($\Delta\rho = \rho_\uparrow - \rho_\downarrow$) isosurface of V-, Cr-, Mn-, Fe- and Co- adsorbed fluorinated borophene as displayed in Fig. 3, provides direct visualization of magnetic zones near adsorbed atoms.

Near the adsorbed transition metal atom, a non-uniform distribution of the spin density difference isosurface has been observed. This is due to the unequal distribution of spin-up and spin-down electrons in the adsorbed atom's d_{xy} , d_{yz} , d_{zx} , d_{xy}^2 and d_{yz}^2 orbitals. The interaction of electrons in the 3d orbital of transition metal atoms with the surrounding fluorine atoms, significantly influence the spin distribution, which results in uneven spin isosurface. The isosurface of Mn adsorption, exhibits a highly symmetric and strongly confined spin cloud surrounding the Mn-atom, which indicates that the Mn-adsorbed structure exhibits maximum unpaired electron for inducing magnetic ordering. The order of spin density difference distribution variation (Mn > Cr > Fe

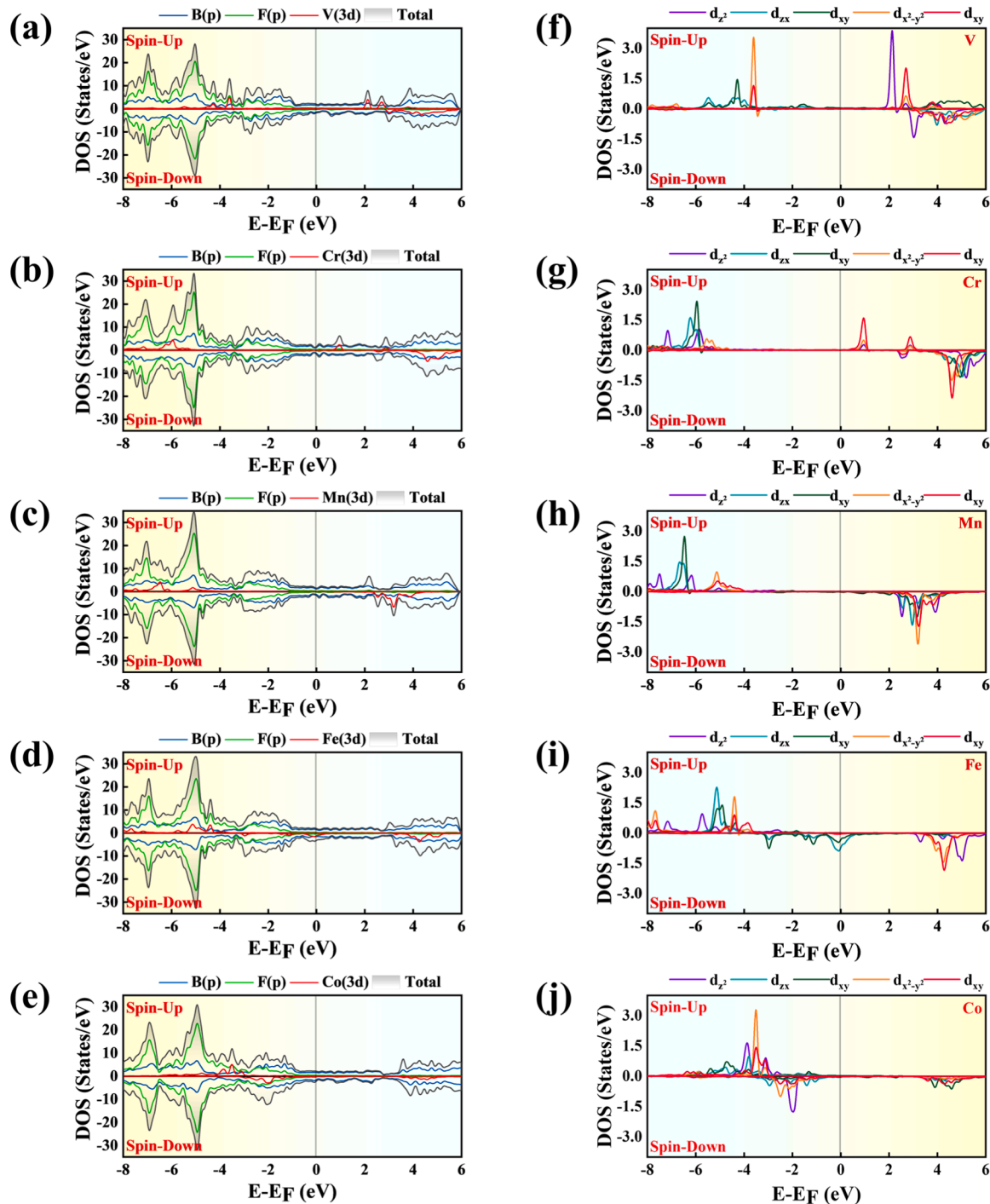


Fig. 4. Partial density of States (PDOS) of (a) V-, (b) Cr-, (c) Mn-, (d) Fe- and (e) Co-adsorbed structure, and from (f) to (j) d_z^2 , d_{xz} , d_{yz} , $d_{x^2-y^2}$ and d_{xy} component of 3d orbital of adsorbed transition metal atom, respectively.

$> \text{Co} > \text{V}$), directly correlates with the magnetic moment and electron distribution in 3d orbital after interacting with surrounding fluorine and boron atoms.

Total density of states with partial density of states (PDOS) of V-, Cr-, Mn-, Fe- and Co-adsorbed fluorinated borophene have been displayed in Fig. 4(a) to (j). The total density of states of transition metal adsorbed fluorinated borophene with boron's 2p-orbital, fluorine's 2p-orbital and transition metal's 3d orbitals have been exhibited in Fig. 4(a) to (e). All the transition metal adsorbed structures exhibit spin splitting between the spin-up and spin-down states, indicating the existence of magnetic exchange interaction, which lead to net magnetic ordering in the adsorbed systems. The PDOS reveal that the 3d orbital of adsorbed transition metal atoms primarily contributed toward splitting in spin-up and spin-down states. For V-, Cr- and Mn-adsorbed structure, spin-up states dominate the PDOS of transition metal's 3d-orbital. This indicates that after adsorption, TM atoms donate 3d-electrons to fluorinated borophene structure and the remaining un-paired electron in the

3d-orbital stabilize to spin-up state. These localized spin-up states are responsible for inducing the magnetic ordering in the structure. The PDOS of 3d orbital of Mn-adsorbed structure exhibit the largest population of occupied spin-up states and high density of un-occupied spin-down states, which give rise to maximum magnetic moment of $4.87 \mu_B$ as compared to other transition metal adsorbed structures. The significant spin-down states have been observed with the spin-up states in 3d orbital of Fe- and Co-adsorbed structures. The partial occupancy of both spin-states decreases the unpaired electron and respective magnetic moment in the system. While all the transition metal adsorbed structures exhibit localized magnetic moments, the Fe-adsorbed structure exhibit more itinerant electronic character, as the spin-down states of Fe contribute significantly at the Fermi level. This indicates stronger hybridization of Fe and fluorinated borophene, however, in the other TM adsorbed structure, the 3d-electronic states remain away from the Fermi level, which is consistent with comparatively localized magnetic behavior [44].

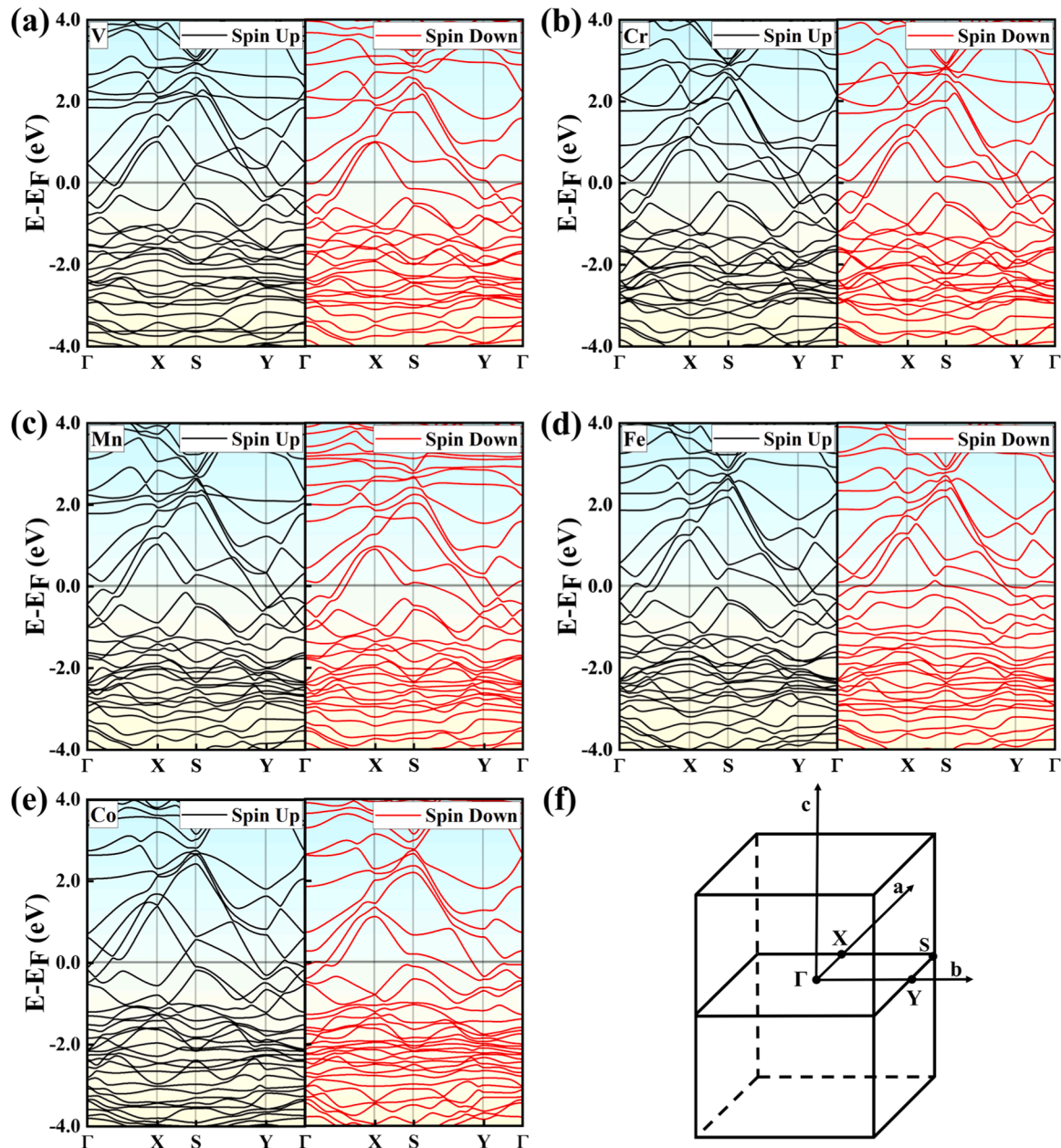


Fig. 5. Electronic band structure of (a) V-, (b) Cr-, (c) Mn-, (d) Fe- and (e) Co-adsorbed structure, and (f) Brillouin zone with high-symmetry path for rectangular two-dimensional structure.

To gain the deeper insight of the orbital contribution to magnetic moment, d_z^2 , d_{zx} , d_{xy} , $d_{x^2-y^2}$ and d_{xy} components of the 3d-orbital of adsorbed transition metal atoms have also been displayed in Fig. 4(f) to (j).

To further unveil the electronic properties of the transition metal adsorbed structure, electronic band structure has been shown in Fig. 5 (a) to (e). The corresponding Brillouin zone and k-path has been displayed in Fig. 5(f). It has been observed that several electronic bands cross the fermi level, which confirm the metallic nature of V-, Cr-, Mn-, Fe- and Co-adsorbed structures. The asymmetry in the band structure of spin-up and spin-down bands confirm the degree of spin polarization while preserving the metallic nature of the fluorinated borophene.

3.4. Degree of spin polarization

The total PDOS reveals that the adsorption of TM atom influences the occupancy of electronic states at fermi level as shown in Fig. 4(a) to (j). To quantify this change, the degree of spin polarization has been estimated by utilizing the following relation [45],

$$P = \frac{N_{\uparrow}(E_F) - N_{\downarrow}(E_F)}{N_{\uparrow}(E_F) + N_{\downarrow}(E_F)} \times 100\% \quad (3a)$$

where, $N_{\uparrow}(E_F)$ and $N_{\downarrow}(E_F)$ represent the density of states at fermi level for spin-up and spin-down states, respectively. The spin polarization is an important feature to evaluate spin-dependent transport behavior, since it offers a numerical representation of the asymmetry between spin-up and spin-down electronic states. The quantified values of degree of spin polarization (P) for V-, Cr-, Mn-, Fe- and Co-adsorbed fluorinated borophene have been displayed in Table 3. Fe-adsorbed γ -B₈F₄ structure exhibits maximum degree of spin-polarization (−22.33%), where the negative sign indicates that spin-down states of 3d orbital of Fe dominate at fermi level. All the d-orbitals in the case of Fe adsorbed structure, except d_{zx} - and d_{xy} - orbitals, remain unoccupied near fermi level as exhibited in Fig. 4(i). The relatively lower degree of spin-polarization has been observed in V-, Cr-, Mn- and Co-adsorbed structures as displayed in Table 3. The minor changes in density of states near fermi level of these structure are due to the electron redistribution in fluorinated borophene structure.

3.5. Magnetic exchange interactions and temperature dependence magnetization

The magnetic exchange interaction plays a key role to stabilize the long-range magnetic ordering in materials. The magnitudes and signs of the exchange coupling constants help to align neighboring spins parallel or antiparallel. Within the Heisenberg model, a positive exchange constant favors parallel spin arrangement and negative exchange constant favors anti-parallel spin arrangement, which corresponds to ferromagnetic and antiferromagnetic coupling, respectively. Based on spin alignment four magnetic configuration, FM, Stripe-y AFM, Stripe-x AFM and Neel's-AFM, has been considered. The possible spin arrangement and exchange interaction pathway has been displayed in Fig. S1. The nature of magnetic ordering in transition metal adsorbed fluorinated borophene has been revealed by the relative magnitudes and signs of the exchange coupling constants J_1 (along x-axis), J_2 (along y-axis), and J_3

(along diagonal). The exchange parameters J_1 , J_2 , and J_3 , have been calculated using Eq. (6) to (8) and displayed in Table 4. The Co-adsorbed fluorinated borophene exhibits ferromagnetic ordering, as all dominant exchange interactions (+ 1.126 meV, + 0.71 meV and + 0.912 meV) are positive, leading to the parallel arrangement. The V-, Cr-, Mn-, and Fe-adsorbed fluorinated borophene exhibit negative exchange parameters, which point out that antiparallel spin alignment is energetically preferred. The presence of multiple competing antiferromagnetic interactions can introduce spin-frustration in the system. This spin frustration arises in the system, when the exchange interactions cannot be simultaneously satisfied by any single collinear spin arrangement. In the rectangular lattice, the negative value of J_1 , J_2 , and J_3 suggest that the diagonal coupling (J_3) competes with the nearest-neighbor interactions, which potentially introduces magnetic frustration. When magnitude J_3 is sufficiently large relative to J_1 and J_2 , the system experiences competing exchange constraints, which can stabilize stripe-type phases, as displayed by V-adsorbed systems.

The pristine fluorinated borophene sheet itself is metallic and non-magnetic, the magnetic interaction between TM atoms is mediated by substrate-induced spin polarization rather than the direct overlap of their 3d orbitals.

In the adsorbed structure, the typical distance between adjacent TM adatoms is approximately 6.28 Å to 6.36 Å, as shown in Table 4, which is relatively large for direct orbital overlap between neighboring magnetic atoms. Under such weak conditions, conventional direct and double exchange is unlikely to dominate. The type of interaction mediated in the adsorbed system closely align with Ruderman–Kittel–Kasuya–Yosida (RKKY)-type exchange interaction. In the metallic systems, which contain dilute localized magnetic moments, the dominant exchange interaction arises from indirect coupling mediated by itinerant conduction electrons, in which magnetic interaction is possessed by the presence of conduction electron of borophene structure. The adsorbed transition metal atom's localized spin moment polarizes the fluorinated borophene sheet's surrounding conduction electrons, generating an oscillatory spin density that extends over long distances. An indirect exchange interaction is formed when this conduction electrons interacts with the nearby magnetic impurity. The exchange constants in Table 4, exhibit noticeable variation in sign and magnitude, which indicate the RKKY-type. Furthermore, compared to three-dimensional hosts, the lower dimensionality of two-dimensional metallic systems like borophene increases the range of the RKKY-type interaction. This low-dimensional effect further increases the possibility long-range magnetic interaction between dilute adatoms. As a result, the fluorinated borophene provides potential platform for creating tunable magnetic phases through controlled adatoms selection.

To further find out the thermodynamics stability of predicted magnetic phase, finite-temperature properties have been calculated utilizing Monte-Carlo simulation based on Heisenberg model. The magnetic phase transition has been observed from the temperature dependence magnetization and specific heat, as shown in Fig. 6. The calculated values of Curie/Neel's temperature have been displayed in Table 4. It

Table 3

Degree of spin polarization with spin up and spin down states at fermi level for of V-, Cr-, Mn-, Fe- and Co- adsorbed fluorinated borophene.

TM atom	$N_{\uparrow}(E_F)$	$N_{\downarrow}(E_F)$	P (%)
V	2.01	1.79	5.79
Cr	01.53	1.74	−6.42
Mn	1.46	1.71	−7.88
Fe	2.0	3.15	−22.33
Co	2.15	2.14	0.23

Table 4

Magnetic exchange interaction parameters, distance between dopant d_{TM-TM} , Magnetic coupling and Curie/Neel's Temperature ($T_{C/N}$) for of V-, Cr-, Mn-, Fe- and Co- adsorbed fluorinated borophene structures.

TM atom	J_1 (meV)	J_2 (meV)	J_3 (meV)	d_{TM-TM} (Å)	Coupling	$T_{C/N}$ (K)
V	−1.189	−1.929	−1.237	6.29	Stripe-x AFM	28
Cr	−0.181	−0.169	−0.051	6.28	Stripe-y AFM	1.59
Mn	−0.038	−0.014	0.069	6.32	Neel's-AFM	1.84
Fe	15.517	−7.39	1.067	6.33	Stripe-x AFM	148.87
Co	1.126	0.71	0.912	6.36	FM	34.5

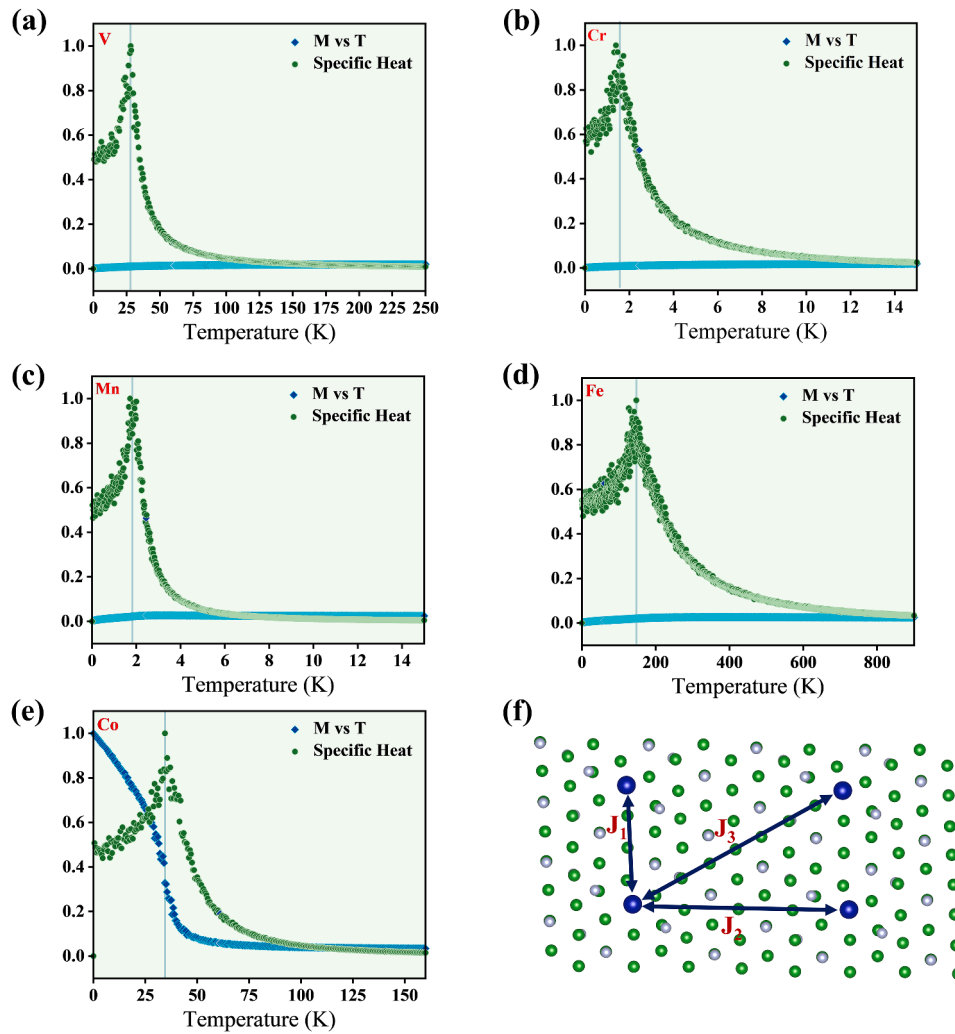


Fig. 6. Magnetization and Specific Heat in normalized unit as function of Temperature for Co-(a)V-, (b) Cr-, (c) Mn-, (d) Fe- and (e) Co-adsorbed fluorinated borophene, (f) Magnetic exchange interaction parameters.

has been found that in Co-adsorbed structure the magnetization decreases with increasing temperature follows up the behavior of FM material and a prominent peak of specific heat has been observed at temperature 34.5 K. The Co-adsorbed fluorinated borophene exhibits a clear transition from ferromagnetic to paramagnetic phase with a Curie temperature of approximately 34.5 K. Although, the V-, Cr-, Mn- and Fe- adsorbed structure undergo antiferromagnetic transitions. The Cr-adsorbed system exhibits the lowest transition temperature (1.59 K), which correlates well with the extremely weak exchange parameters. Within the Heisenberg model, the critical temperature scales approximately with the effective exchange strength. Therefore, weak exchange interaction leads to enhanced thermal spin fluctuations at lower temperature. On the other hand, the highest transition temperature (148.87 K) in the Fe-adsorbed system results from the strong ferromagnetic coupling along one direction and antiferromagnetic coupling along the opposite direction, which increase the overall energy scale of magnetic ordering. This correlation of magnetic ordering with exchange interaction strength provides strong foundation for designing practical 2D magnetic devices.

3.6. Magnetic anisotropy energy (MAE)

Long-range magnetic ordering in two-dimensional materials is stabilized by Magnetic Anisotropy Energy (MAE), particularly when the magnetic moment is intensely localized on a transition-metal atom. The difference between the energies corresponding to the magnetization

direction in the in-plane ($E_{||}$) and out-plane ($E_{\perp}$) is referred to Magnetic anisotropy energy ($MAE = E_{||} - E_{\perp}$). Since our results indicate that Mn-adsorbed γ -B₈F₄ exhibits the maximum magnetization (4.87 μ_B) among all transition metal adsorbed γ -B₈F₄ structure, assessing its MAE provides vital information about the robustness of its magnetic behavior. The force field theorem has been used to calculate the magnetic anisotropy energy, and the calculated result exhibits an MAE value of 1.681 meV for the Mn-adsorbed structure. The positive value of MAE confirms that Mn-adsorbed γ -B₈F₄ exhibits out-of-plane easy-axis magnetization, which stabilizes the magnetism in two-dimensional system by suppressing thermal spin fluctuations and overcomes the limitation provided by Mermin-Wagner theorem [23].

4. Discussion

The findings of this study reveal the emergence of magnetic ordering in 3d transition-metal adsorbed fluorinated borophene (γ -B₈F₄) and highlight the interplay between chemical functionalization, electron correlation and exchange mechanisms with importance of magnetic anisotropy energy (MAE) to stabilize the magnetic ordering in two-dimensional materials. The emergence of magnetism in these systems is due to the partially filled 3d orbitals of the adsorbed transition-metal atoms. The electronic configuration of the isolated TM atom, the degree of hybridization with the B- and F-derived states, and the on-site electron correlation considered by the Hubbard U correction, affect the

magnetic moment as shown in Table 2.

Among all the adsorbed TM atoms, Mn adsorbed γ -B₈F₄ structure exhibits the largest magnetic moment of 4.87 μ_B , which is consistent with its half-filled 3d⁵ configuration in isolated form. The projected density of states (PDOS) exhibited in Fig. 4(c) and (h), reveals that 3d-orbital of Mn has a significant exchange splitting between the majority and minority spin states, which identify that most of the minority-spin states are empty. This strong spin polarization reduces the orbital quenching and results in a compact, symmetric spin-density isosurface around the Mn atom, as observed in the isosurface plots in Fig. 3(c). On the other hand, Co- and Fe- adsorbed γ -B₈F₄ structures exhibit more complex 3d occupancies and display reduced magnetic moments due to a higher degree of hybridization with the fluorinated borophene. Fe and Co adsorbed γ -B₈F₄ structure have partially occupied minority spin-states. The PDOS of Fe adsorbed γ -B₈F₄ structure reveals that minority spin states intersect the Fermi level, indicating itinerant magnetism with delocalized spin states extending into the host structure as shown in Fig. 4(f). This delocalization influences the nature of magnetic coupling and reduces the net moment.

An equally important factor in stabilizing magnetism in two-dimensional systems is magnetic anisotropy energy (MAE), which determines the favorable direction for spin alignment. The non-collinear spin polarized calculation reveals that Mn- adsorbed γ -B₈F₄ exhibit a significant out-of-plane MAE of 1.681 meV. The d_z^2 , d_{zx} , d_{xy} , $d_{x^2-y^2}$ and d_{xy} component of 3d orbital in Mn-adsorbed structure experience strong directional asymmetry due to the orbital interaction with surrounding atoms. This findings is highly relevant because two-dimensional material lacking magnetic anisotropy cannot sustain magnetic ordering at finite temperature due to thermal fluctuations [23]. This finding thus reveals TM-adsorbed fluorinated borophene as two-dimensional magnetic material for potential application in spin memory, data storage and magnetic device application.

5. Conclusion

In conclusion, we have demonstrated that fluorination stabilizes 2-Pmmn borophene efficiently, and that a robust magnetic ordering has been introduced by the subsequent adsorption of 3d transition metal atoms utilizing first-principles DFT+*U* simulations. All of the TM atoms (V, Cr, Mn, Fe, and Co) adsorbed structure exhibit negative formation energies, suggesting thermodynamic stability. All the TM adsorbed structures exhibit localized magnetic moment except Fe, which exhibit notable 3d-orbital's density of states near Fermi level. While the Co-adsorbed structure exhibit ferromagnetic ordering having Curie temperature (34.5 K) The Mn-adsorbed structure exhibit highest magnetic moment (4.87 μ_B) and a significant out-of-plane magnetic anisotropy energy (1.681 meV), demonstrating that magnetism in two-dimensional material is strongly stabilized by anisotropy. It has been observed that RKKY-Type of interaction stabilize long range ferromagnetic ordering in the TM adsorbed structures. This work provides substantial insight into the correlation between chemical modification and magnetism in 2D materials, in addition to establishing fluorinated borophene as a suitable platform for magnetic functionalization. Our research opens the door to the development of high T_C, stable, and controllable magnetic material for application in spintronic devices.

CRediT authorship contribution statement

Paras Poswal: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Neeraj Shukla:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.surf.2026.109143.

Data availability

Data will be made available on request.

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